

## NGARI, China

WMO: 552280

Lat: 32.500N

Lon: 80.083E

Elev: 4280

StdP: 59.42

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |             |              |            |              | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|-------------|--------------|------------|--------------|-----------------------|-------------|------------|-------------|-----------------------|-----------|--------------|
|               |              |              | 99.6%                         |            |             | 99%          |            |              | 0.4%                  |             | 1%         |             |                       |           |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB        | DP           | HR         | MCDB         | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-24.0 | (c)<br>-20.9 | (d)<br>-37.6                  | (e)<br>0.2 | (f)<br>-7.9 | (g)<br>-35.7 | (h)<br>0.2 | (i)<br>-10.1 | (j)<br>11.2           | (k)<br>-2.9 | (l)<br>9.8 | (m)<br>-3.7 | (n)<br>0.3            | (o)<br>70 | (p)<br>0.710 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |             |            |             |            |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|-------------|------------|-------------|------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |             | 1%         |             | 2%         |             |                         |            |
|               |                           | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB        | WB         | MCDB        | WB         | MCDB        | MCWS                    | PCWD       |
| (a)<br>7      | (b)<br>11.7               | (c)<br>22.7     | (d)<br>7.2 | (e)<br>21.6 | (f)<br>7.1 | (g)<br>20.5 | (h)<br>6.9 | (i)<br>9.6          | (j)<br>18.2 | (k)<br>8.9 | (l)<br>17.2 | (m)<br>8.3 | (n)<br>16.6 | (o)<br>4.3              | (p)<br>250 |

| Dehumidification DP/MCDB and HR |      |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 6.7                             | 10.4 | 12.1 | 5.9 | 9.9 | 12.0 | 5.1 | 9.3 | 12.3 | 41.7          | 18.2 | 39.6 | 17.3 | 37.8 | 16.6 | 14.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 9.6           | (b) 8.3 | (c) 7.3 | (e) -26.5                  | (f) 25.0 | (g) 3.7            | (h) 1.1 | (i) -29.2                                          | (j) 25.8 | (k) -31.4  | (l) 26.4 | (m) -33.5  | (n) 27.0 | (o) -36.2  | (p) 27.8 |    |    |
| (a) 9.6           | (b) 8.3 | (c) 7.3 | (e) -27.2                  | (f) 10.5 | (g) 3.3            | (h) 1.1 | (i) -29.6                                          | (j) 11.4 | (k) -31.6  | (l) 12.0 | (m) -33.4  | (n) 12.6 | (o) -35.8  | (p) 13.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 1.7    | -11.4 | -8.9  | -4.6  | 0.6   | 5.4   | 11.1  | 15.0  | 14.2  | 9.7   | 1.1   | -4.4  | -8.5  |  |  |
|                                                                        | DBStd       | 9.47   | 4.13  | 3.75  | 3.32  | 2.76  | 2.66  | 3.20  | 1.79  | 1.84  | 2.91  | 3.14  | 2.64  | 3.63  |  |  |
|                                                                        | HDD10.0     | 3418   | 662   | 530   | 452   | 282   | 145   | 27    | 1     | 0     | 41    | 276   | 432   | 573   |  |  |
|                                                                        | HDD18.3     | 6083   | 921   | 763   | 710   | 532   | 402   | 217   | 103   | 129   | 258   | 534   | 682   | 831   |  |  |
|                                                                        | CDD10.0     | 377    | 0     | 0     | 0     | 0     | 1     | 59    | 156   | 129   | 32    | 0     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 15     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 3     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.6    | 2.1   | 2.6   | 2.8   | 2.9   | 3.1   | 3.1   | 3.0   | 2.6   | 2.5   | 2.1   | 1.9   | 2.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 67     | 2     | 1     | 2     | 1     | 2     | 3     | 21    | 26    | 6     | 1     | 1     | 1     |  |  |
|                                                                        | PrecMax     | 136    | 6     | 4     | 11    | 6     | 14    | 17    | 50    | 58    | 27    | 9     | 11    | 9     |  |  |
|                                                                        | PrecMin     | 21     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 25     | 2     | 1     | 2     | 1     | 3     | 4     | 13    | 16    | 7     | 2     | 2     | 2     |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 2.1   | 4.0   | 9.6   | 13.0  | 17.3  | 22.8  | 24.7  | 23.7  | 20.7  | 14.7  | 8.1   | 4.9   |  |  |
|                                                                        |             | MCWB   | -5.4  | -4.1  | -1.8  | 0.0   | 3.3   | 6.2   | 8.6   | 8.0   | 6.2   | 1.0   | -2.6  | -4.5  |  |  |
|                                                                        | 2%          | DB     | -0.4  | 1.7   | 6.7   | 11.0  | 15.6  | 21.4  | 23.3  | 22.0  | 19.1  | 12.4  | 6.0   | 2.3   |  |  |
|                                                                        |             | MCWB   | -6.9  | -5.4  | -2.8  | -0.7  | 2.1   | 6.1   | 7.5   | 7.5   | 5.6   | 0.4   | -3.8  | -5.7  |  |  |
|                                                                        | 5%          | DB     | -2.3  | -0.2  | 4.6   | 9.5   | 14.1  | 20.0  | 22.2  | 20.8  | 17.7  | 10.6  | 4.4   | 0.4   |  |  |
|                                                                        |             | MCWB   | -8.0  | -6.6  | -4.0  | -1.4  | 1.5   | 5.8   | 7.3   | 7.2   | 5.1   | -0.4  | -4.5  | -6.7  |  |  |
|                                                                        | 10%         | DB     | -4.1  | -1.8  | 2.9   | 8.1   | 12.7  | 18.5  | 21.0  | 19.7  | 16.4  | 8.8   | 2.8   | -1.4  |  |  |
|                                                                        |             | MCWB   | -9.2  | -7.6  | -4.9  | -2.0  | 0.8   | 5.1   | 7.1   | 7.2   | 4.4   | -1.4  | -5.2  | -7.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -4.9  | -3.2  | -1.5  | 0.5   | 4.3   | 8.5   | 11.5  | 10.6  | 7.8   | 2.7   | -2.4  | -3.9  |  |  |
|                                                                        |             | MCDB   | 0.9   | 3.0   | 8.1   | 10.6  | 14.8  | 19.0  | 21.3  | 19.3  | 17.4  | 11.4  | 7.0   | 3.8   |  |  |
|                                                                        | 2%          | WB     | -6.5  | -5.1  | -2.6  | -0.3  | 3.2   | 7.5   | 9.9   | 9.6   | 7.0   | 1.1   | -3.4  | -5.5  |  |  |
|                                                                        |             | MCDB   | -1.3  | 0.9   | 6.0   | 9.8   | 13.0  | 18.0  | 18.8  | 17.8  | 16.2  | 10.8  | 5.3   | 1.7   |  |  |
|                                                                        | 5%          | WB     | -7.8  | -6.3  | -3.7  | -1.0  | 2.3   | 6.7   | 9.1   | 9.0   | 6.3   | -0.1  | -4.3  | -6.6  |  |  |
|                                                                        |             | MCDB   | -2.7  | -0.7  | 4.0   | 8.5   | 11.7  | 17.1  | 17.7  | 16.9  | 15.2  | 9.4   | 3.9   | 0.0   |  |  |
|                                                                        | 10%         | WB     | -9.0  | -7.4  | -4.7  | -1.7  | 1.4   | 5.9   | 8.5   | 8.5   | 5.6   | -1.2  | -5.2  | -7.6  |  |  |
|                                                                        |             | MCDB   | -4.2  | -2.1  | 2.5   | 7.3   | 10.9  | 16.2  | 17.1  | 16.2  | 14.0  | 8.2   | 2.6   | -1.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.1  | 12.5  | 13.1  | 13.3  | 13.3  | 12.6  | 11.7  | 10.8  | 12.1  | 13.7  | 14.8  | 14.1  |  |  |
|                                                                        |             | MCDBR  | 14.5  | 14.6  | 15.6  | 15.5  | 15.0  | 13.9  | 13.5  | 12.2  | 12.9  | 15.0  | 16.3  | 15.6  |  |  |
|                                                                        |             | MCWBR  | 10.1  | 9.9   | 9.3   | 7.9   | 6.7   | 5.2   | 4.6   | 3.6   | 4.7   | 7.4   | 9.7   | 10.3  |  |  |
|                                                                        | 5% WB       | MCDBR  | 13.5  | 13.6  | 14.5  | 14.2  | 13.4  | 12.6  | 11.2  | 10.3  | 11.2  | 13.8  | 15.7  | 15.3  |  |  |
|                                                                        |             | MCWBR  | 9.6   | 9.3   | 8.9   | 7.4   | 6.2   | 4.9   | 4.0   | 3.5   | 4.5   | 7.0   | 9.5   | 10.2  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.195 | 0.221 | 0.255 | 0.300 | 0.312 | 0.302 | 0.308 | 0.303 | 0.242 | 0.206 | 0.185 | 0.180 |  |  |
|                                                                        | taud        |        | 2.554 | 2.454 | 2.364 | 2.246 | 2.255 | 2.375 | 2.419 | 2.440 | 2.583 | 2.611 | 2.654 | 2.654 |  |  |
|                                                                        | Ebn at Noon |        | 1070  | 1057  | 1032  | 987   | 972   | 976   | 968   | 973   | 1034  | 1066  | 1080  | 1081  |  |  |
|                                                                        | Edh at Noon |        | 78    | 97    | 116   | 138   | 138   | 122   | 116   | 112   | 92    | 82    | 70    | 67    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 3.40  | 4.22  | 5.85  | 6.85  | 7.54  | 7.60  | 7.03  | 6.47  | 6.26  | 5.57  | 4.38  | 3.60  |  |  |
|                                                                        | RadStd      |        | 0.46  | 0.48  | 0.29  | 0.32  | 0.51  | 0.46  | 0.54  | 0.49  | 0.47  | 0.29  | 0.18  | 0.21  |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (0 neighbors) | N/A   | N/A     | -1.41  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page